

9.3.8

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Question

Find the area of the region $\{(x, y) : x^2 \leq y \leq x\}$.

Theoretical Solution

Let us solve the given equation theoretically and then verify the solution computationally

According to the question,
equation line

$$\mathbf{x} = \lambda \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (1)$$

equation of parabola is

$$\mathbf{x}^T \mathbf{v} \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f = 0 \quad (2)$$

where

$$\mathbf{v} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad (3)$$

$$\mathbf{u} = \begin{pmatrix} 0 \\ \frac{-1}{2} \end{pmatrix} \quad (4)$$

$$f = 0 \quad (5)$$

Theoretical Solution

substitute 0.1 in 0.2

$\lambda = 0, 1$ so intersection points are

$$\mathbf{a}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (6)$$

$$\mathbf{a}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (7)$$

hence the required area is

$$\int_0^1 (x - x^2) dx = \frac{1}{6} \quad (8)$$

```
#include <stdio.h>

int main() {
    // Conic:  $y = x^2$ 
    double V[2][2] = {{1, 0}, {0, 0}};
    double u[2] = {0, -0.5};
    double f = 0;

    // Line:  $y = x$ 
    double n[2] = {1, -1};
    double c = 0;

    // Intersection points
    double P1[2] = {0, 0};
    double P2[2] = {1, 1};

    printf("# Parabola (conic form)\n");
```

```
printf("V = [[%lf, %lf], [%lf, %lf]]\n", V[0][0], V[0][1], V[1][0], V[1][1]);
printf("u = [[%lf], [%lf]]\n", u[0], u[1]);
printf("f = %lf\n", f);

printf("\n# Line (normal form)\n");
printf("n = [[%lf], [%lf]]\n", n[0], n[1]);
printf("c = %lf\n", c);

printf("\n# Intersection Points\n");
printf("P1 = [[%lf], [%lf]]\n", P1[0], P1[1]);
printf("P2 = [[%lf], [%lf]]\n", P2[0], P2[1]);

return 0;
}
```

python Code

```
# Code to plot  $y = x$  and  $y = x^2$  and their intersection points
# Based on GVV Sharma's structure
# Corrected and updated: Sep 30, 2025
```

```
import numpy as np
import matplotlib.pyplot as plt
import sys
```

```
# Path to custom geometry modules
sys.path.insert(0, '/sdcard/github/matgeo/codes/CoordGeo')
```

```
# Local imports (from your repo)
from line.funcs import line_norm
from conics.funcs import conic_param, parab_param, parab_gen
```

```
# Termux (Android) support
import subprocess
import shlex
```

```
# Define unit vectors (if needed)
e1 = np.array([[1], [0]])
e2 = np.array([[0], [1]])

# Set up plot
fig = plt.figure()
ax = fig.add_subplot(111, aspect='equal')

# Conic form:  $x^T V x + 2u^T x + f = 0$ 
V = np.array([[1, 0], [0, 0]]) #  $x^2$  term
u = np.array([[0], [-1/2]]) #  $-y$  term  $2u^T x = -y$ 
f = 0 # constant term

# Get conic parameters using your functions
n, c, F, O, lam, P, e = conic_param(V, u, f)
```

```
# Generate parabola in standard form, then apply affine transform
flen = parab_param(lam, P, u)
y_vals = np.linspace(-2, 2, 100)
x_parab = parab_gen(y_vals, flen)
x_parab_affine = P @ x_parab + 0

# Plot parabola
plt.plot(x_parab_affine[0,:], x_parab_affine[1,:], label='$y = x^2$', color='red')

# Line in normal vector form:  $(1, -1)^T x = 0$ 
n_line = np.array([[1], [-1]])
c_line = 0
x_line = line_norm(n_line, c_line, -2, 2)
```

```
# Plot line
plt.plot(x_line[0,:], x_line[1,:], label='$y = x$', color='blue')
P1 = np.array([[0], [0]])
P2 = np.array([[1], [1]])
intersections = np.hstack((P1, P2))

# Plot intersection points
plt.scatter(intersections[0,:], intersections[1:], color='black',
            , zorder=5)
labels = ['(0, 0)', '(1, 1)']
for i, txt in enumerate(labels):
    plt.annotate(txt,
                 (intersections[0,i], intersections[1,i]),
                 textcoords="offset points",
                 xytext=(0,10), ha='center')
```

```
# Axes through origin
ax.spines['top'].set_color('none')
ax.spines['right'].set_color('none')
ax.spines['left'].set_position('zero')
ax.spines['bottom'].set_position('zero')

# Labels, legend, grid
plt.xlabel('$x$')
plt.ylabel('$y$')
plt.title('Graphs of  $y = x^2$  and  $y = x$  with Intersections')
plt.legend()
plt.grid(True)
plt.axis('equal')
plt.savefig('chapters/11/11/2/2/figs/line_parabola_intersections.
pdf')
subprocess.run(shlex.split("termux-open chapters/11/11/2/2/figs/
line_parabola_intersections.pdf"))

# For desktop use, uncomment:
# plt.show()
```

Graphs of $y = x^2$ (positive branch) and $y = x$ with Intersections

